

Math 290

Practice Quiz 2 — Version B

(extra practice, Sections 6–11)

1. Short response.

(a) Evaluate the following proposed theorem and proof.

Theorem: Given $x \in \mathbb{Z}$, if $2 \mid x$, then x is even.

Proof. Let $x \in \mathbb{Z}$. Assume, contrapositively, that x is even. Then $x = 2k$ for some $k \in \mathbb{Z}$. Hence $2 \mid x$. $\square$

- A. The theorem is true and the proof is correct.
- B. The theorem is false.
- C. The theorem is true, but the proof is incorrect because $2 \mid x$ does not mean that $x = 2k$ for some $k \in \mathbb{Z}$.
- D. The theorem is true, but the proof is incorrect because it proves the converse rather than the contrapositive.
- E. The theorem is true, but the proof is incorrect because $x = 2k$ does not imply that $2 \mid x$.

(b) True or false: The converse of the open sentence “If x is even, then x^2 is even.” is the open sentence “If x is odd, then x^2 is odd.”

(c) True or false: The statement “For any $x \in \mathbb{N}$, if $x \in \emptyset$, then $x > 0$.” is both vacuously true and trivially true.

2. Prove that for any $n \in \mathbb{Z}$, we have $3 \mid n$ if and only if $3 \mid n^2$.

3. Prove that for any $x, y \in \mathbb{Z}$, we have $33x + 132y \neq 57$.

4. Prove the following statement (giving a *single* justification for every step you make).
For any two sets A and B , we have $A \cup B = B$ if and only if $A \subseteq B$.